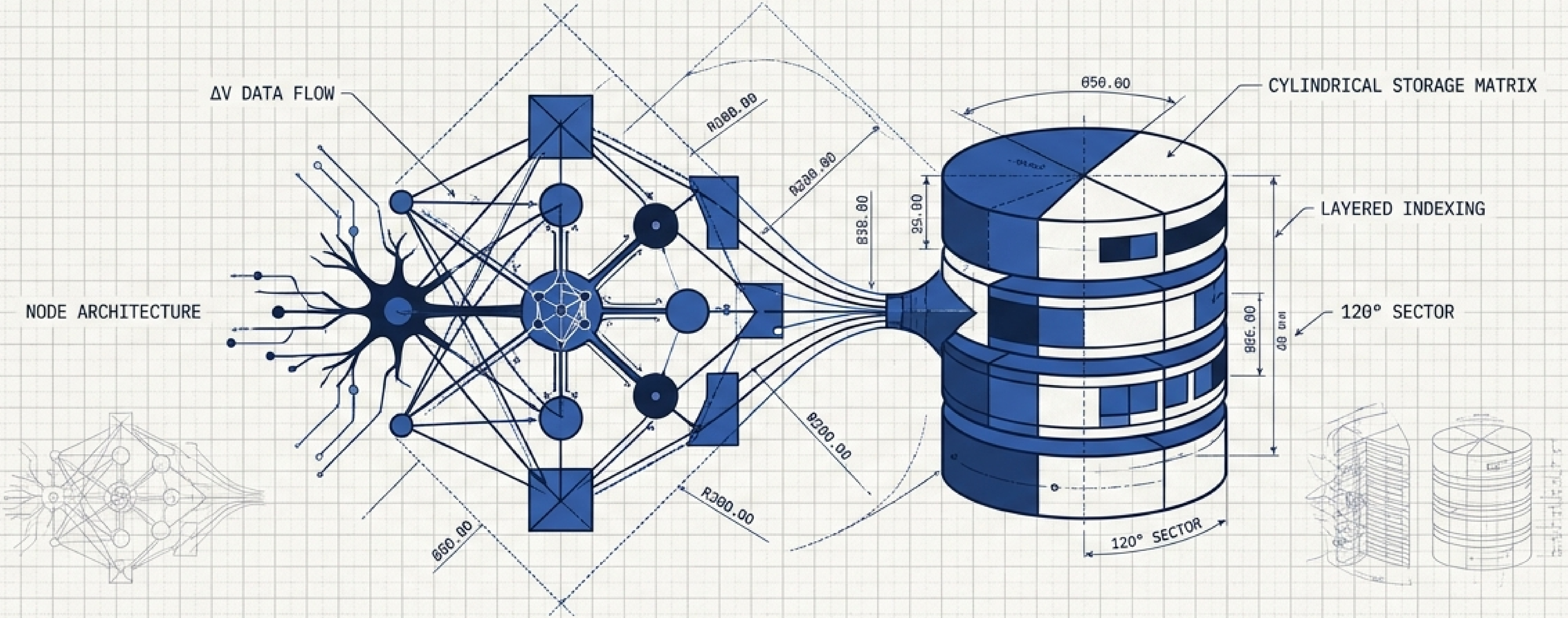
ATLAS OF AI MEMORY SYSTEMS



Architectural Schematics for Agent State and Knowledge Retrieval

A Field Guide to Storage, Ontologies, and Context Engines

COGNEE

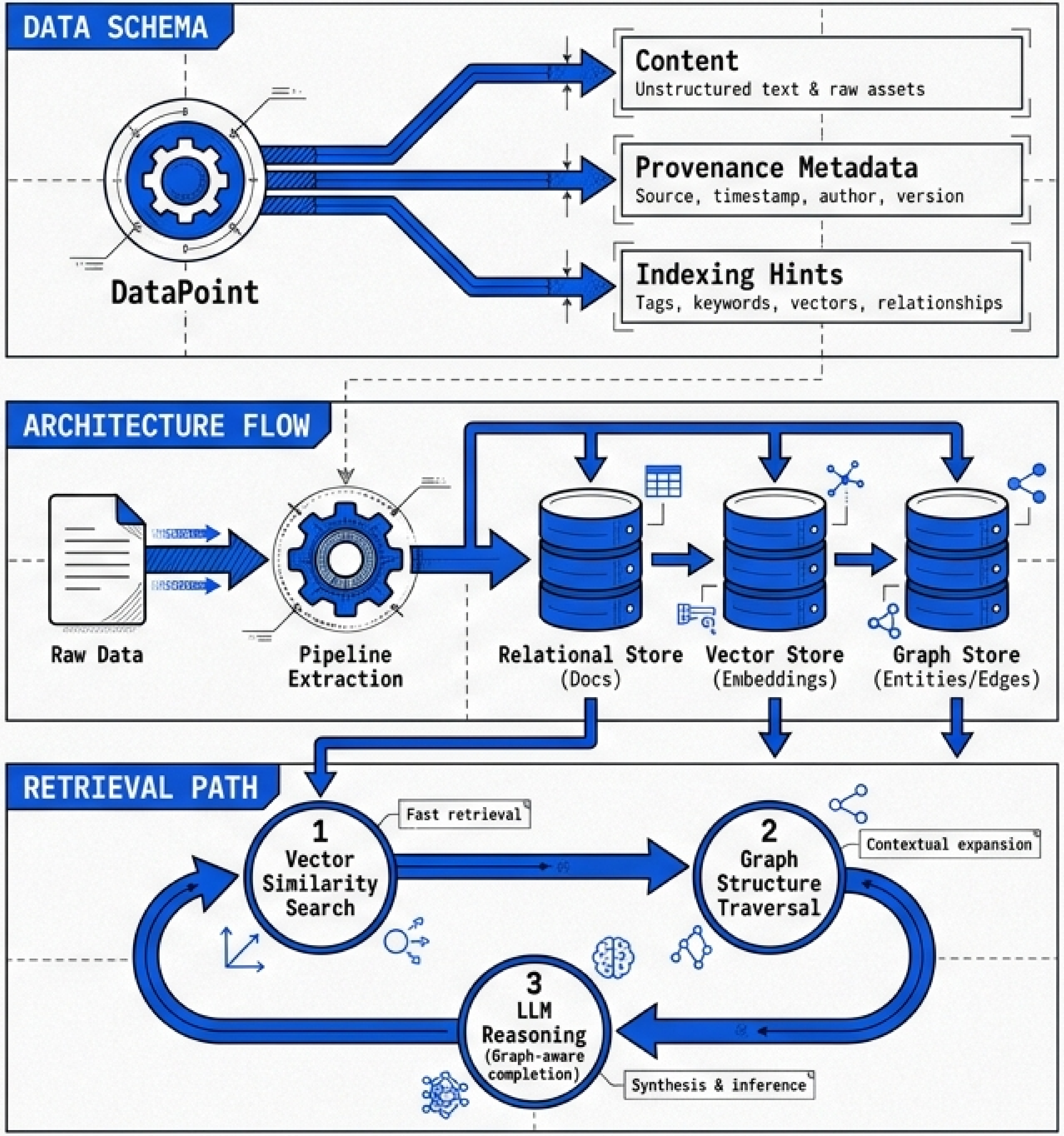
Graph-Emergent Memory System

TARGET VECTOR

Raw data chunks lack relational context.
Documents mention entities, but semantic retrieval alone cannot map how objects relate.

SIGNATURE MECHANISM

Graph generation from unstructured data pipelines via **cognify()**.



MINDBRAIN

TECHNICAL

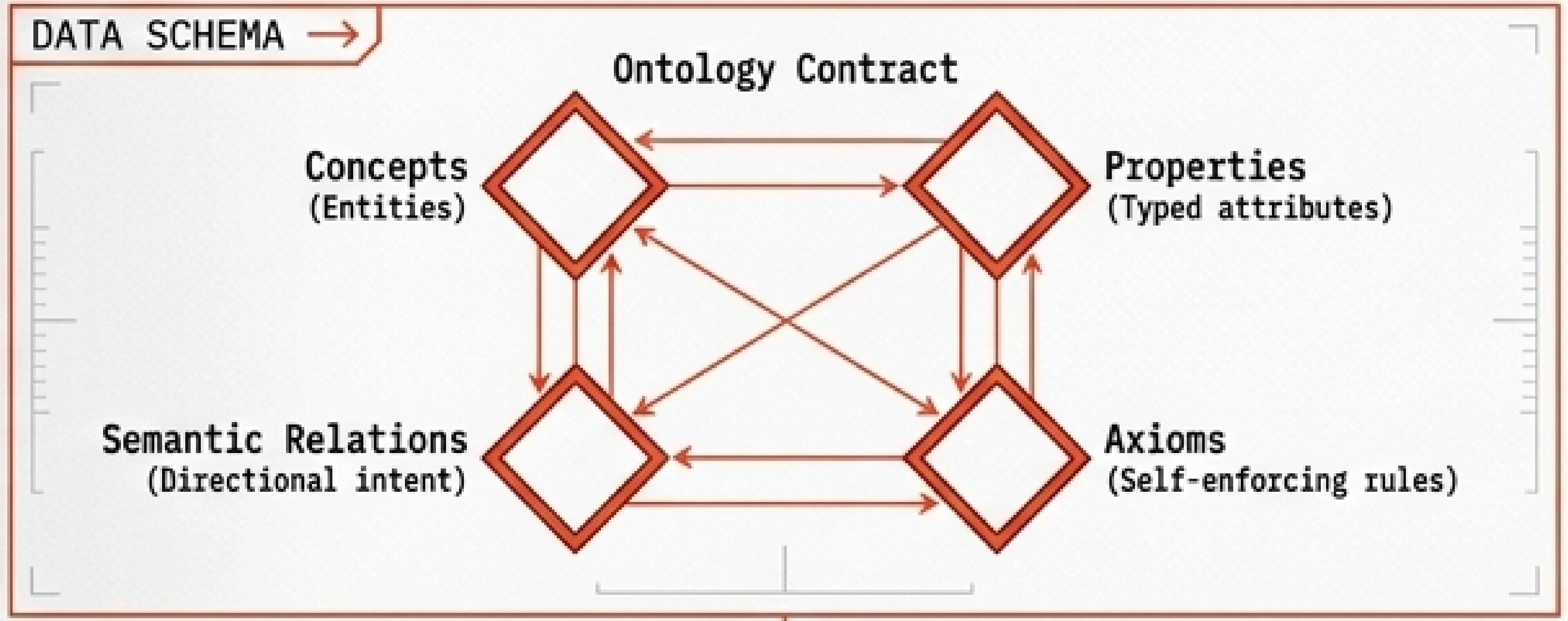
Structured Agentic Database

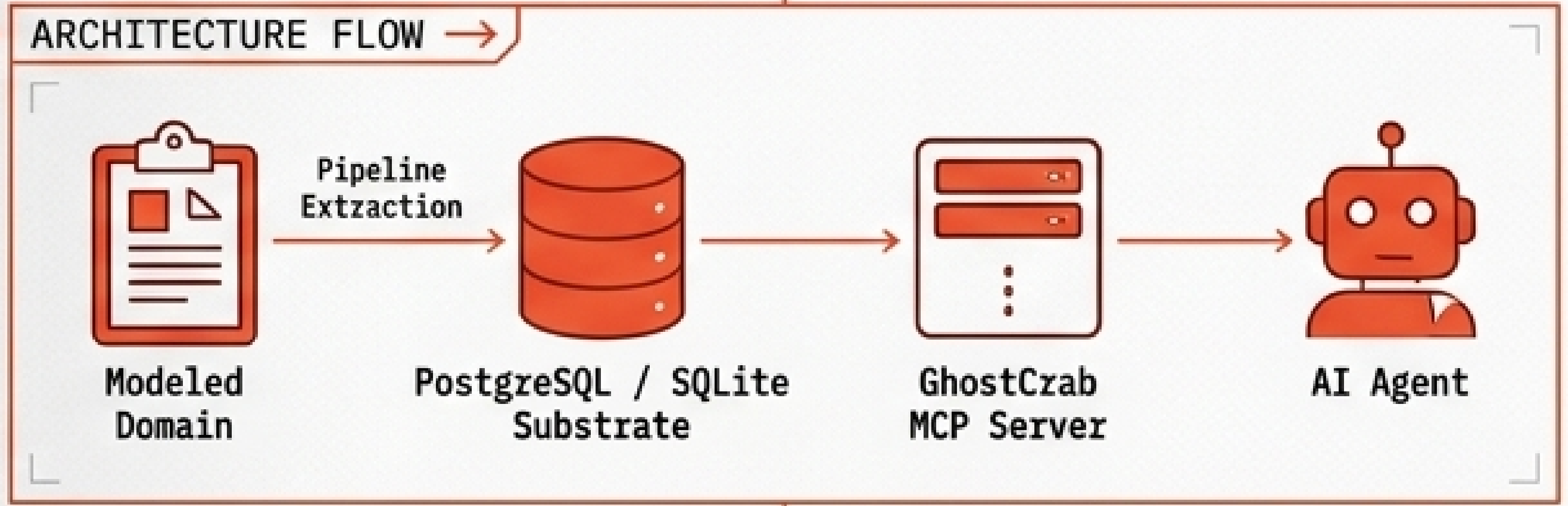
TARGET VECTOR

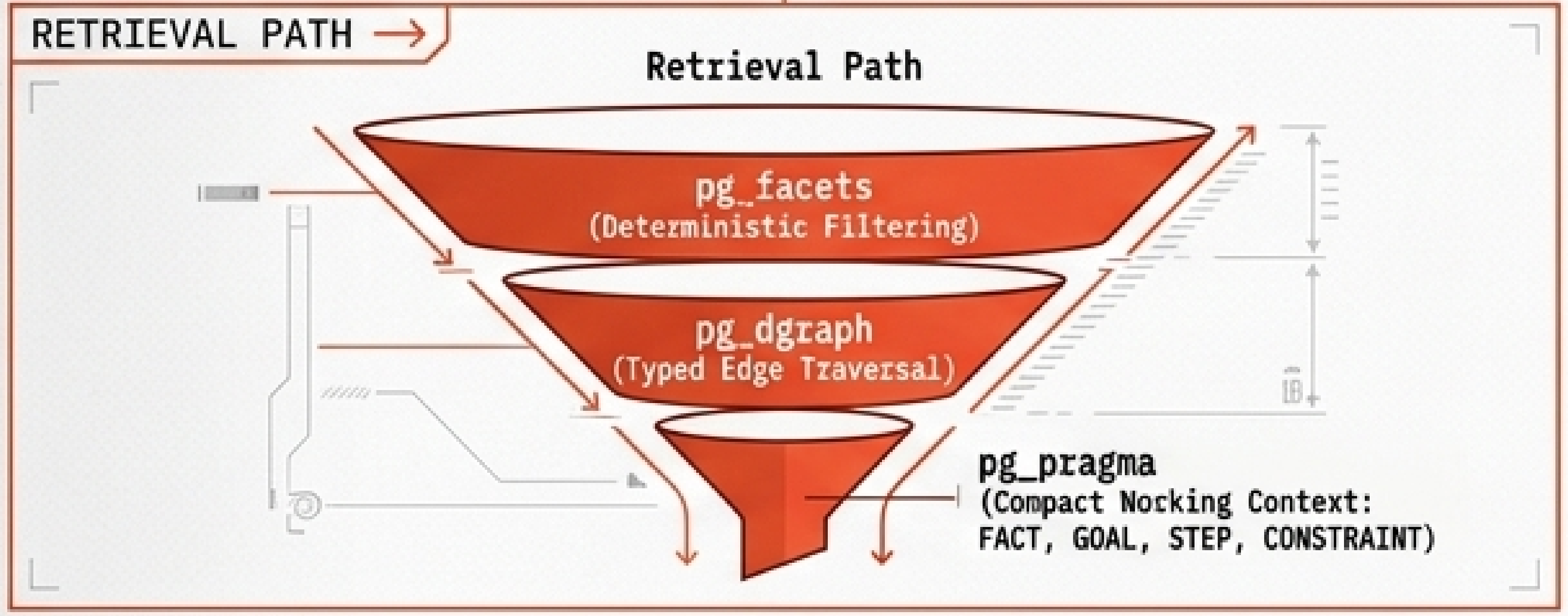
Agents lack durable project state. Unstructured prose cannot reliably enforce dependencies, lifecycle transitions, or compliance blockers.

SIGNATURE MECHANISM

GhostCrab MCP surface exposing Find, Follow, and Pack operations over typed ontologies.







NotebookLM

GBRAIN

TECHNICAL

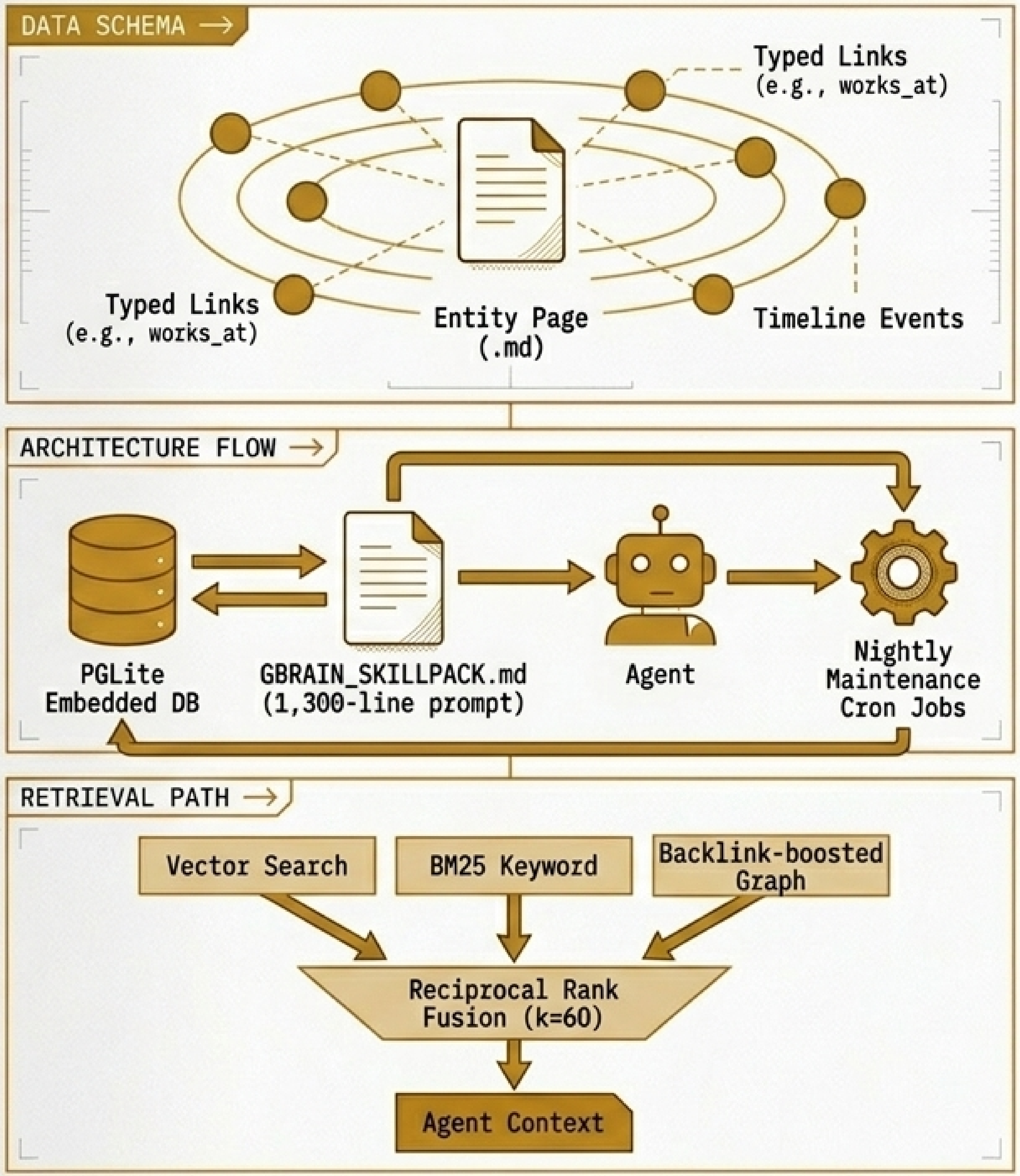
Self-Wiring Personal Knowledge Graph

TARGET VECTOR

Continuous personal knowledge dissipates without manual, high-friction graph wiring and entity maintenance.

SIGNATURE MECHANISM

Nightly ‘Dream Cycle’ for autonomous ingest, repair, enrichment, and consolidation.



HERMES AGENT MEMORY

TECHNICAL

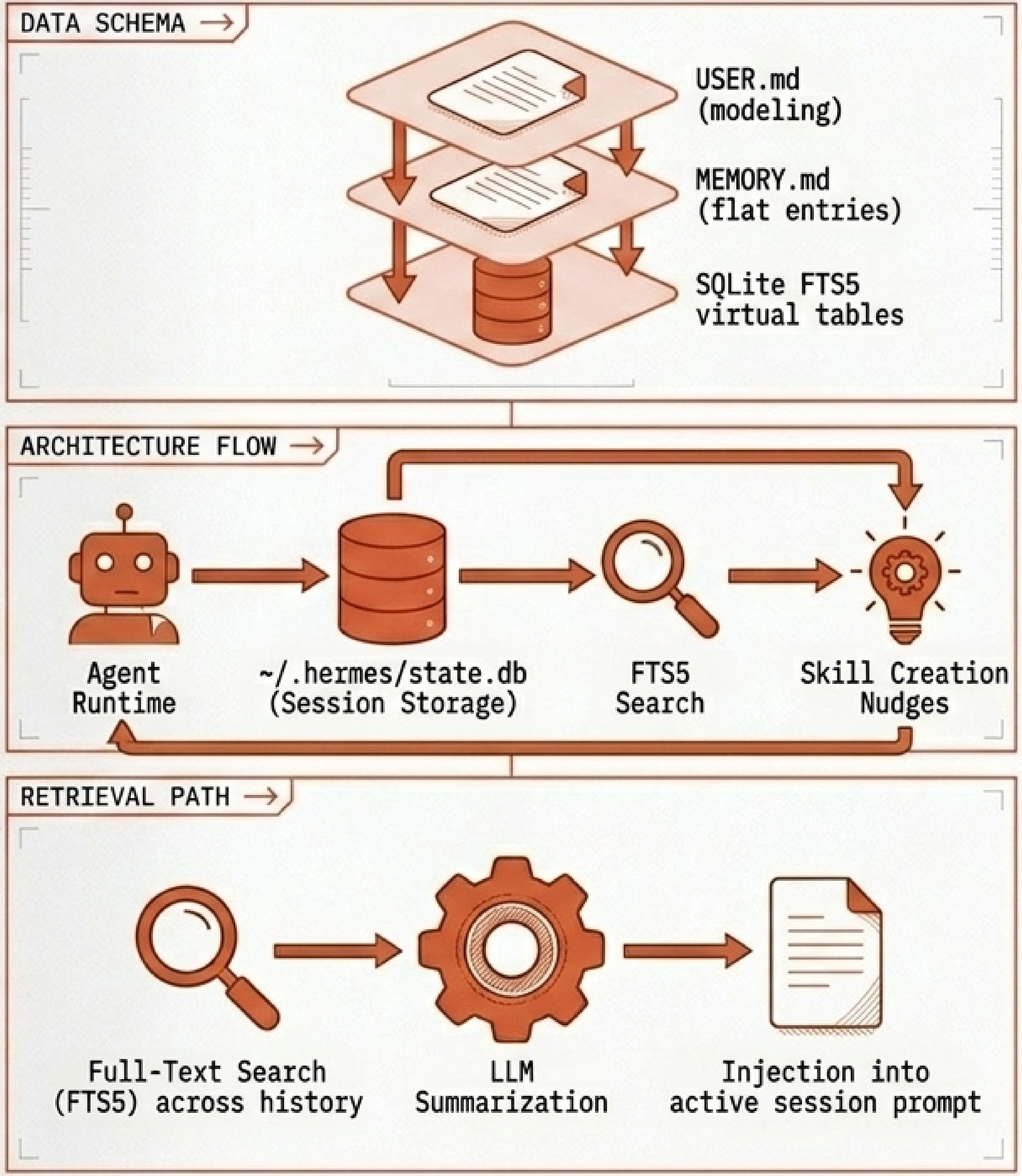
Agent-Native Cross-Session Recall

TARGET VECTOR

Agent sessions are ephemeral. Runtimes cannot preserve user preferences, environments, or reusable skills across sessions.

SIGNATURE MECHANISM

Closed learning loop driven by user modeling and autonomous skill creation.



HINDSIGHT

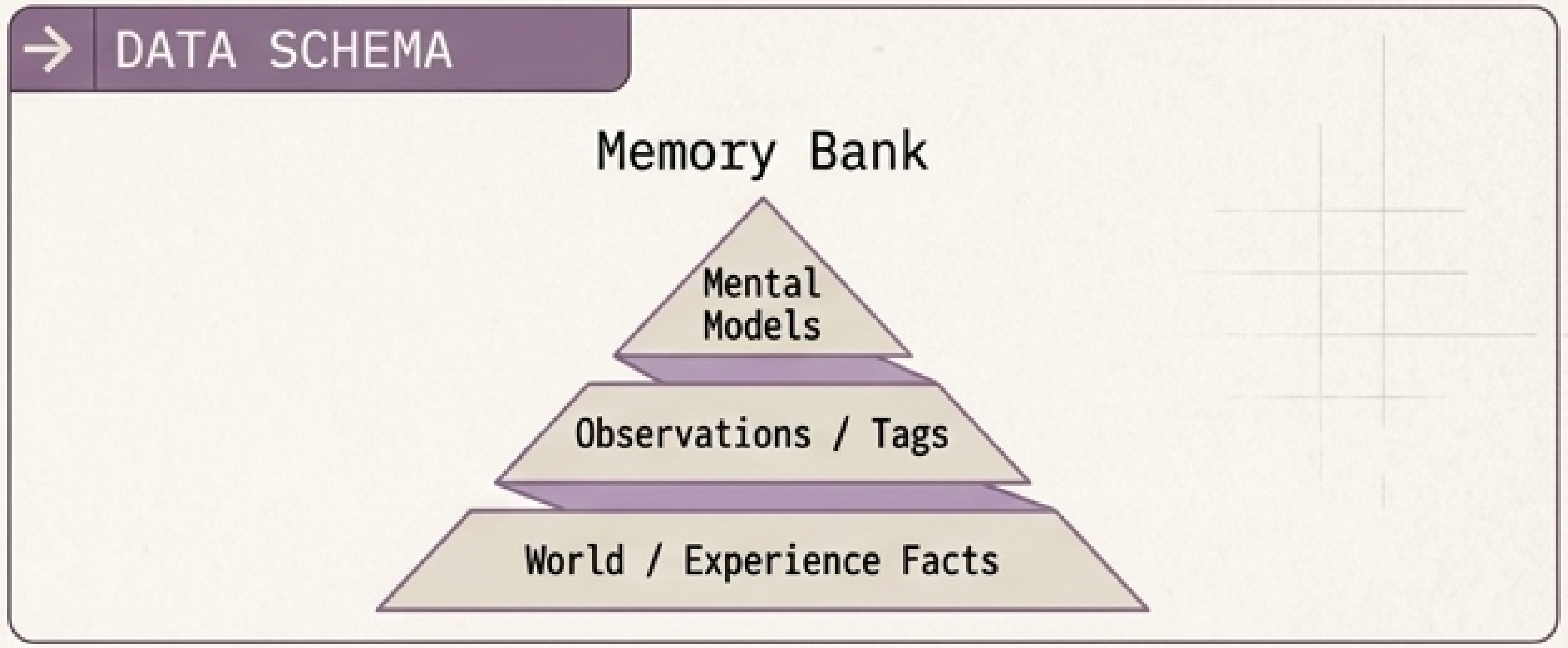
Learning-Oriented Agent Memory System

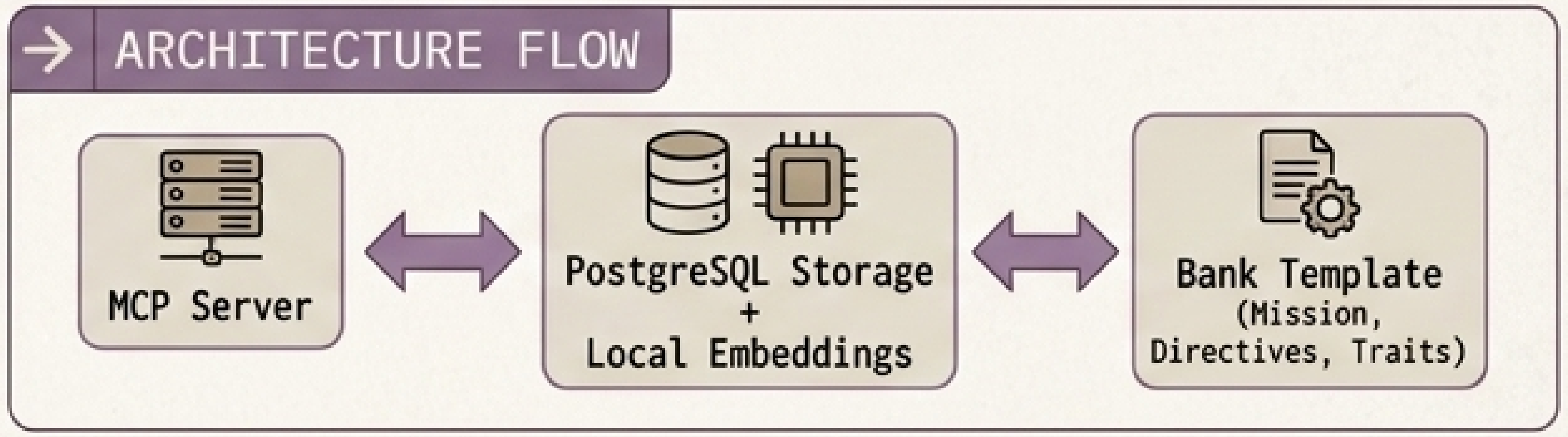
TARGET VECTOR

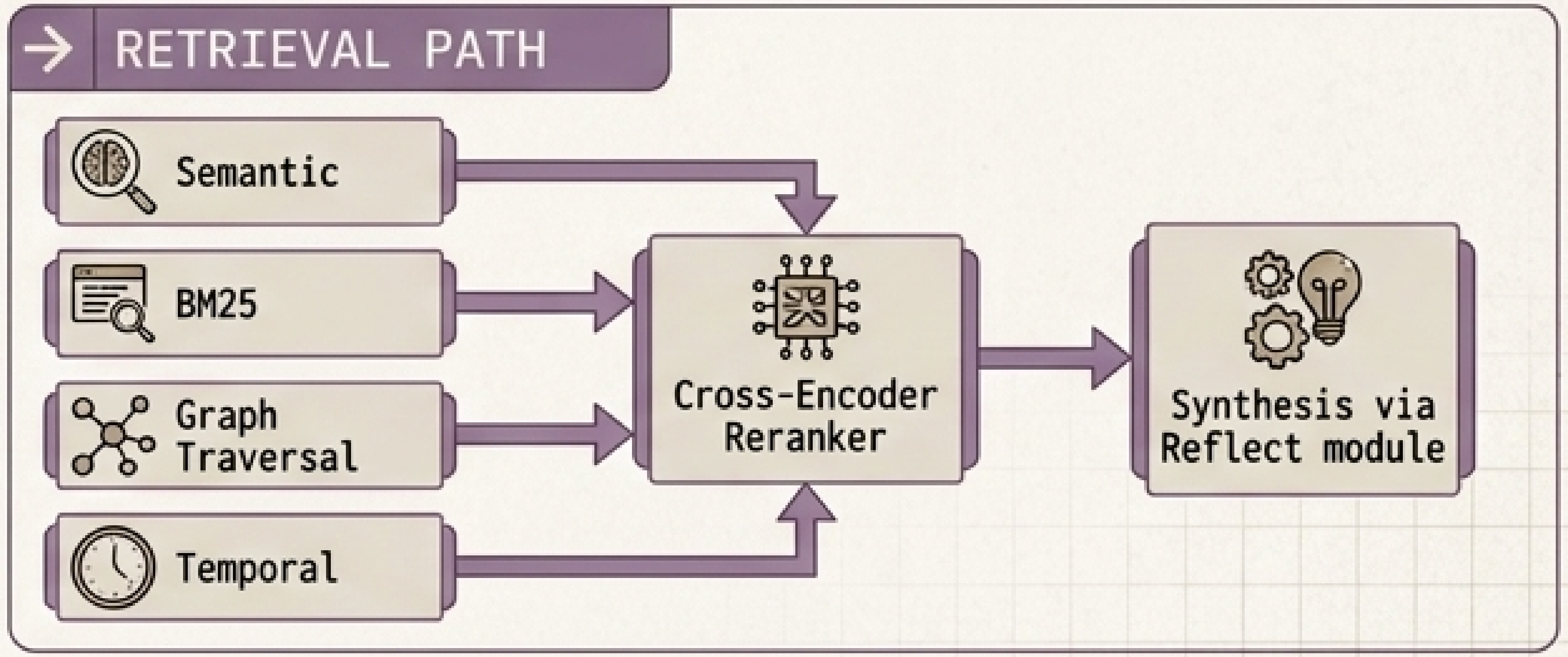
Agents fail to synthesize observed patterns over long horizons, requiring explicit memory extraction and reflection.

SIGNATURE MECHANISM

The 3-phase Retain, Recall, Reflect operation cycle over dedicated Memory Banks.







LANGCHAIN MEMORY

Composable Memory Toolkit

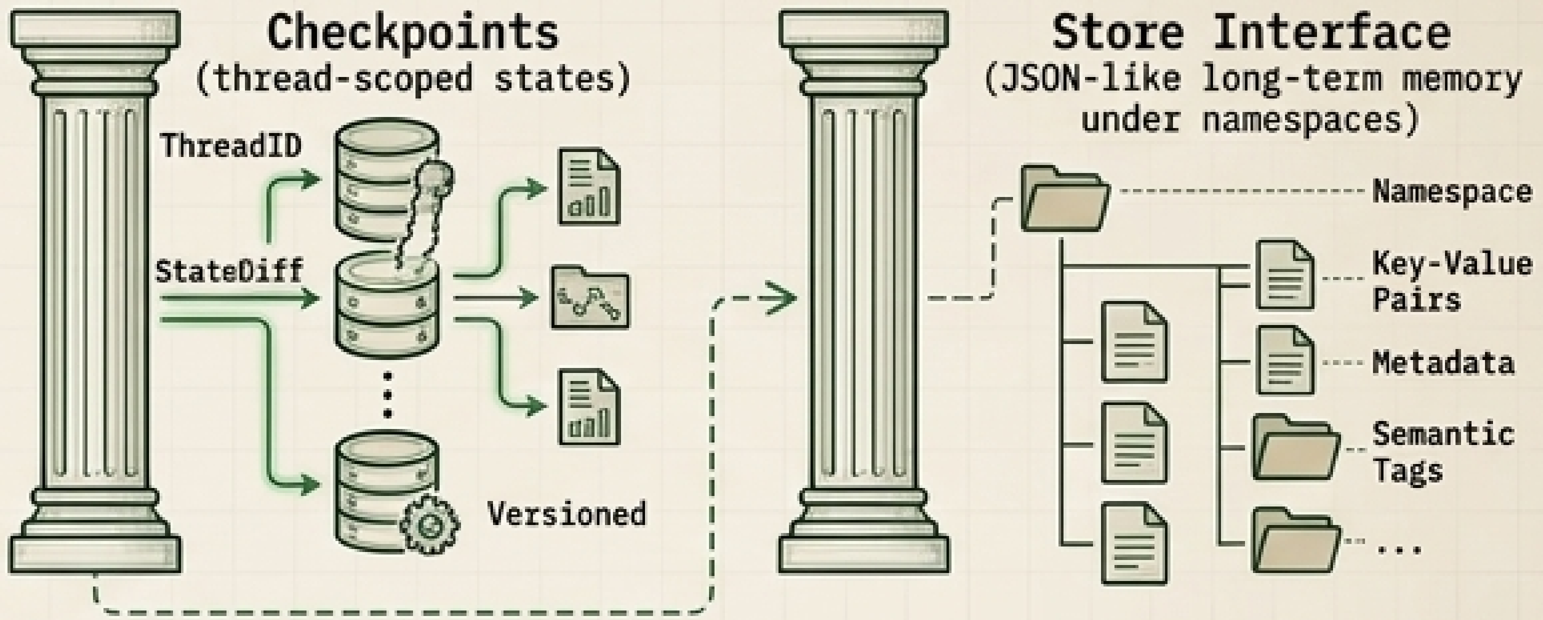
TARGET VECTOR

Workflows require highly customized state persistence, from immediate conversational continuity to thread resumption.

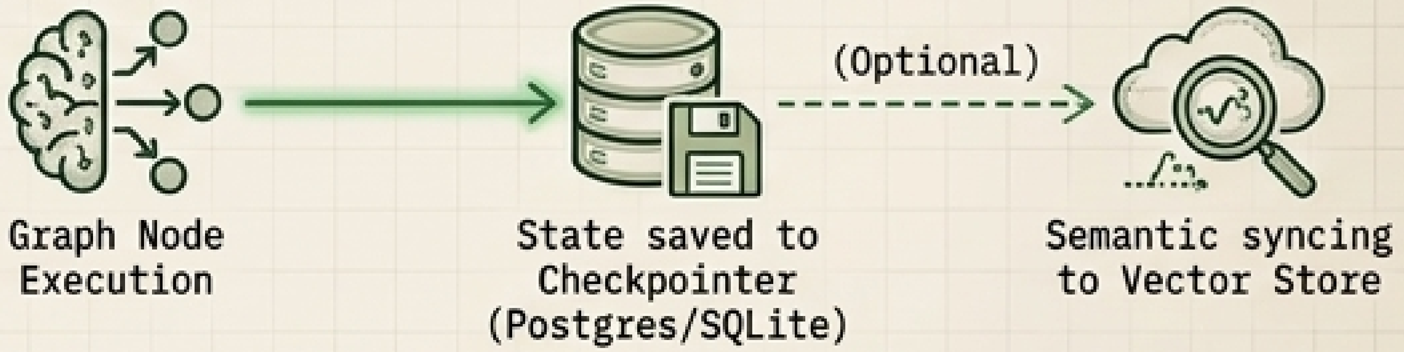
SIGNATURE MECHANISM

Application-controlled memory composition and explicit checkpointer policies.

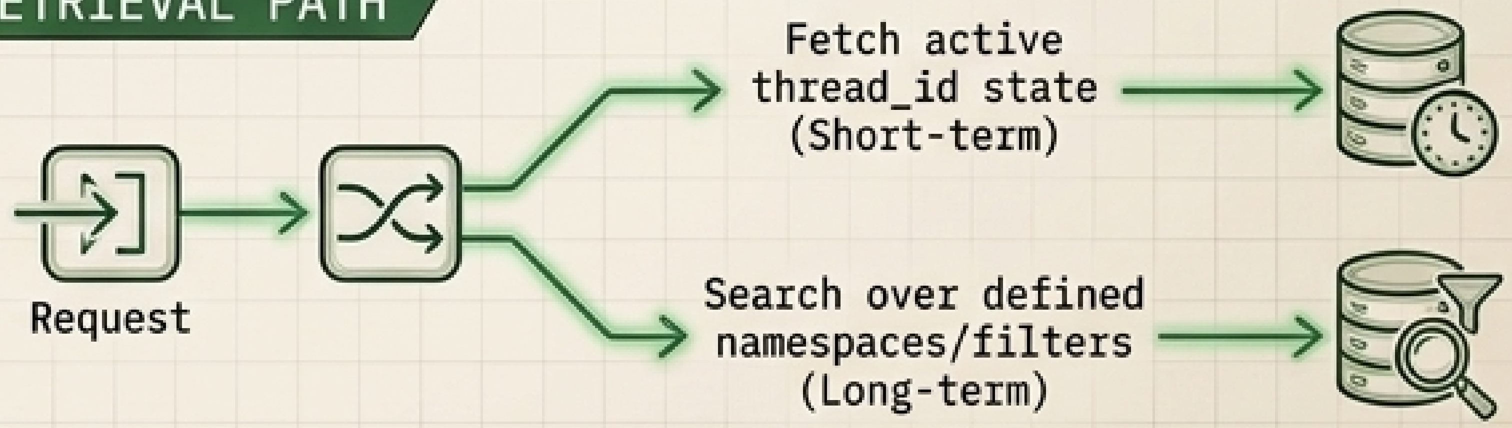
DATA SCHEMA



ARCHITECTURE FLOW



RETRIEVAL PATH



LETTA

Stateful Agent Operating System

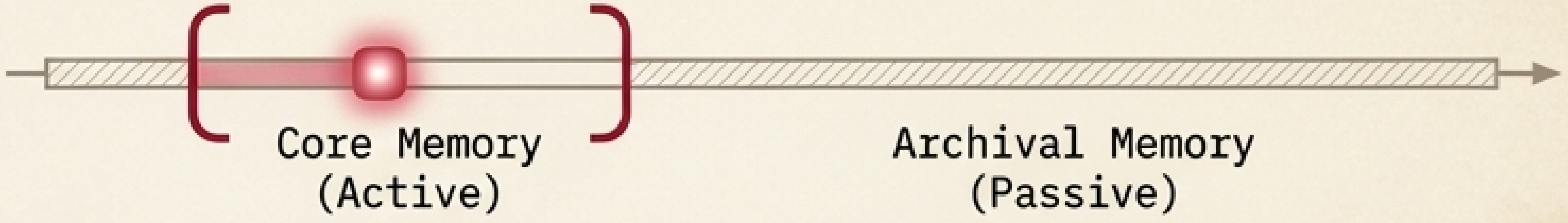
TARGET VECTOR

Prompt-wrapper agents lack agency over their context windows, leading to context overflow or forgotten instructions.

SIGNATURE MECHANISM

Agent-controlled memory editing via explicit runtime tool calls.

DATA SCHEMA



ARCHITECTURE FLOW

Agent Runtime Core ⇄ System Tool Execution (Add/Edit/Search) ⇄ Persistent Storage Layer

RETRIEVAL PATH

Agent detects context gap → Invokes search tool over Archival Memory → Retrieves specific blocks → Writes relevant context into Core Memory limit

LLAMAINDEX

Document and Query Framework

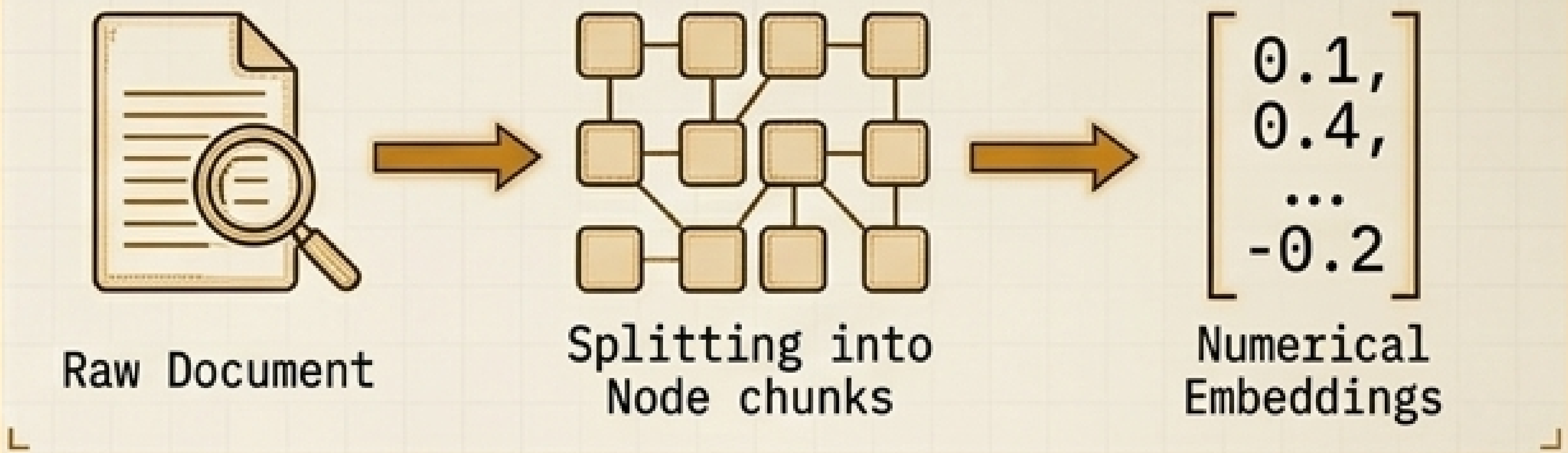
TARGET VECTOR

High-value context is locked inside massive document corpora that exceed context limits and require semantic distillation.

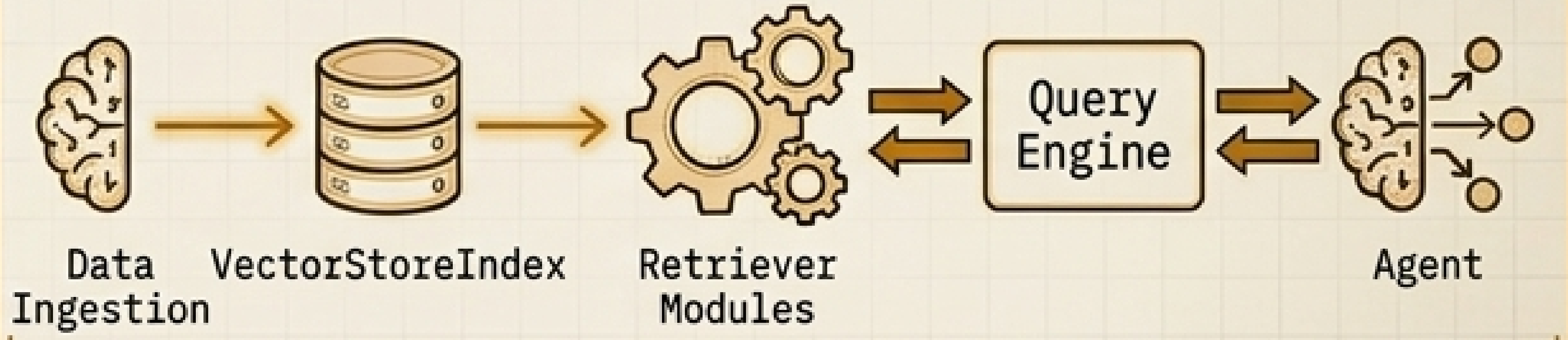
SIGNATURE MECHANISM

Retrieval-centered context construction and token-budget memory flushing.

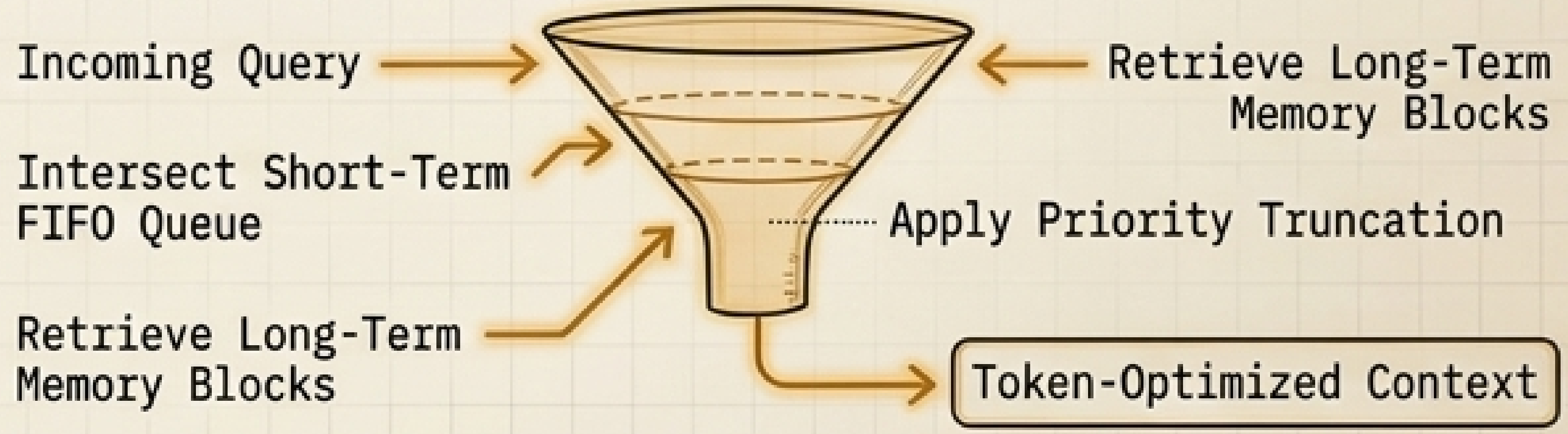
DATA SCHEMA



ARCHITECTURE FLOW



RETRIEVAL PATH



MEM0

Adaptive Memory Infrastructure Layer

TARGET VECTOR

Developers need a frictionless API to add durable user personalization without designing a domain ontology upfront.

SIGNATURE MECHANISM

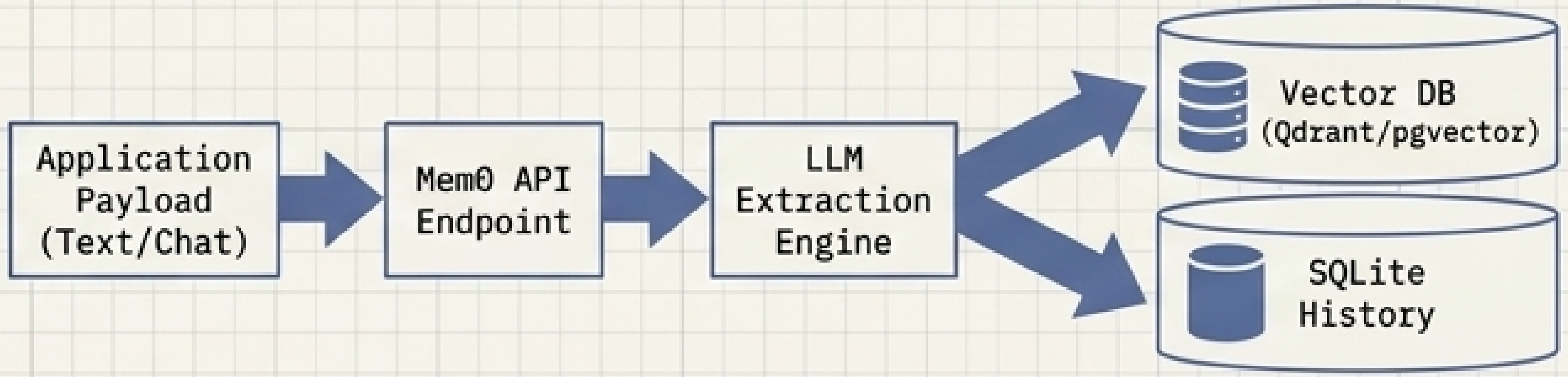
Single-pass ADD-only extraction and hybrid ranking behind a unified developer API.

DATA SCHEMA

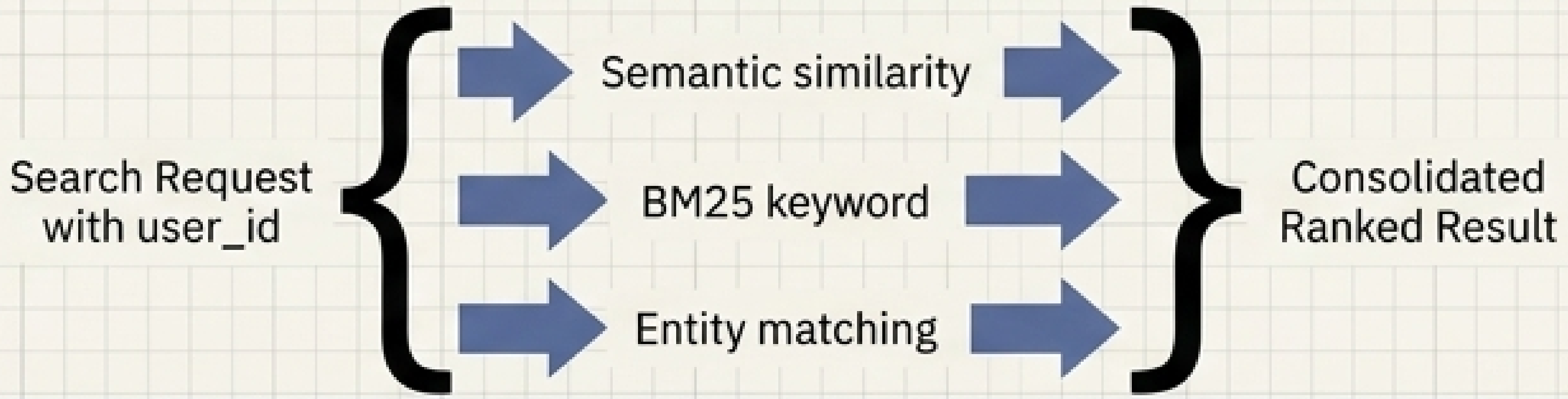
Memory Item Card

ID: [UUID]
Text: [Content Payload]
User Scope: [User ID/Group ID]
Categories: [Tags/Labels]
Timestamp: [ISO 8601 Date]
Relevance Score: [0.0 - 1.0 Float]

ARCHITECTURE FLOW



RETRIEVAL PATH



OBSIDIAN AI (LLM WIKI PATTERN)

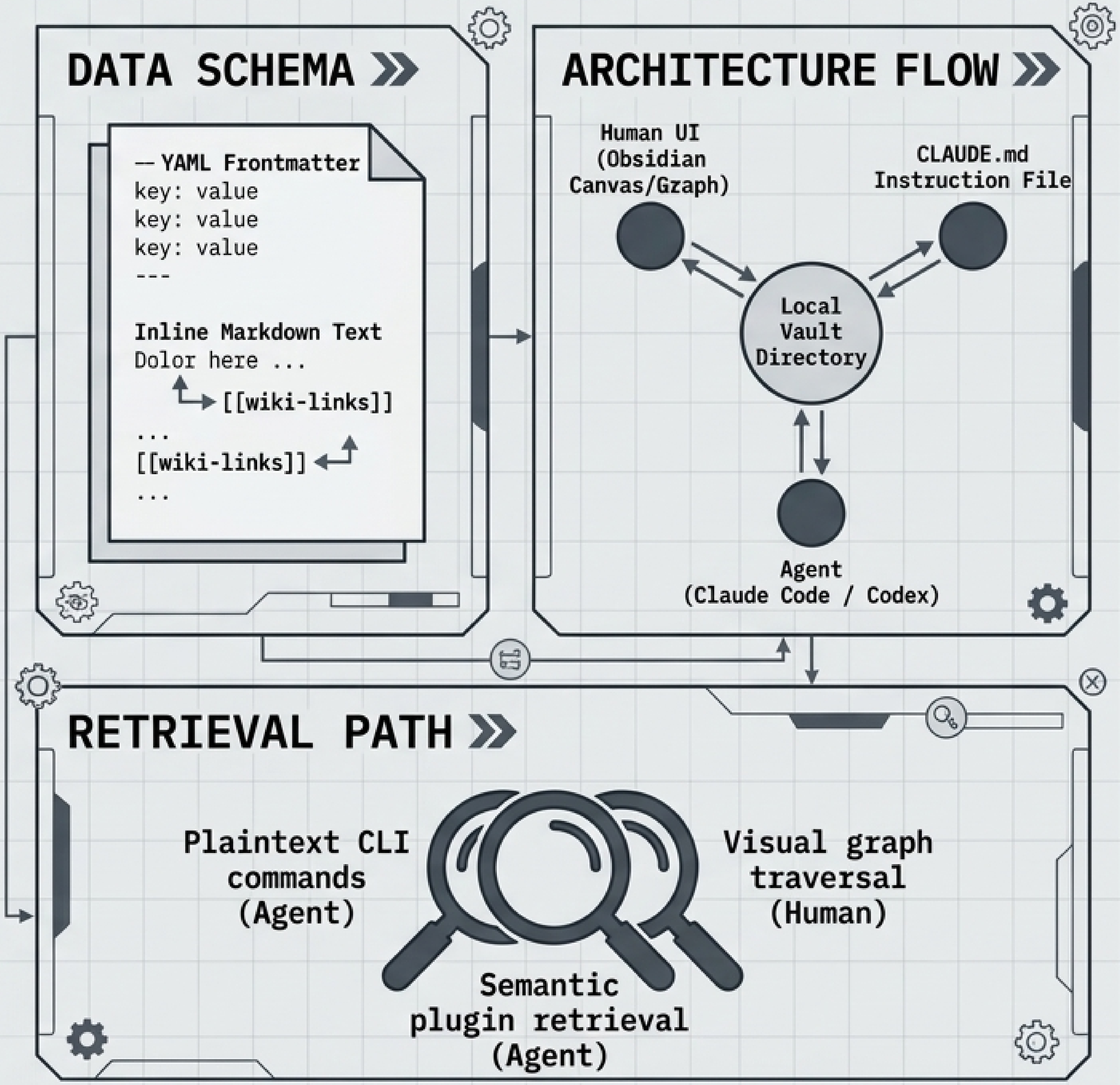
Human-Authored PKM Workspace

TARGET VECTOR »

Opaque vector databases prevent human auditing, editing, and absolute ownership of the knowledge artifact.

SIGNATURE MECHANISM

Agentic construction via direct local file system manipulation under human-readable protocols.



PROTÉGÉ

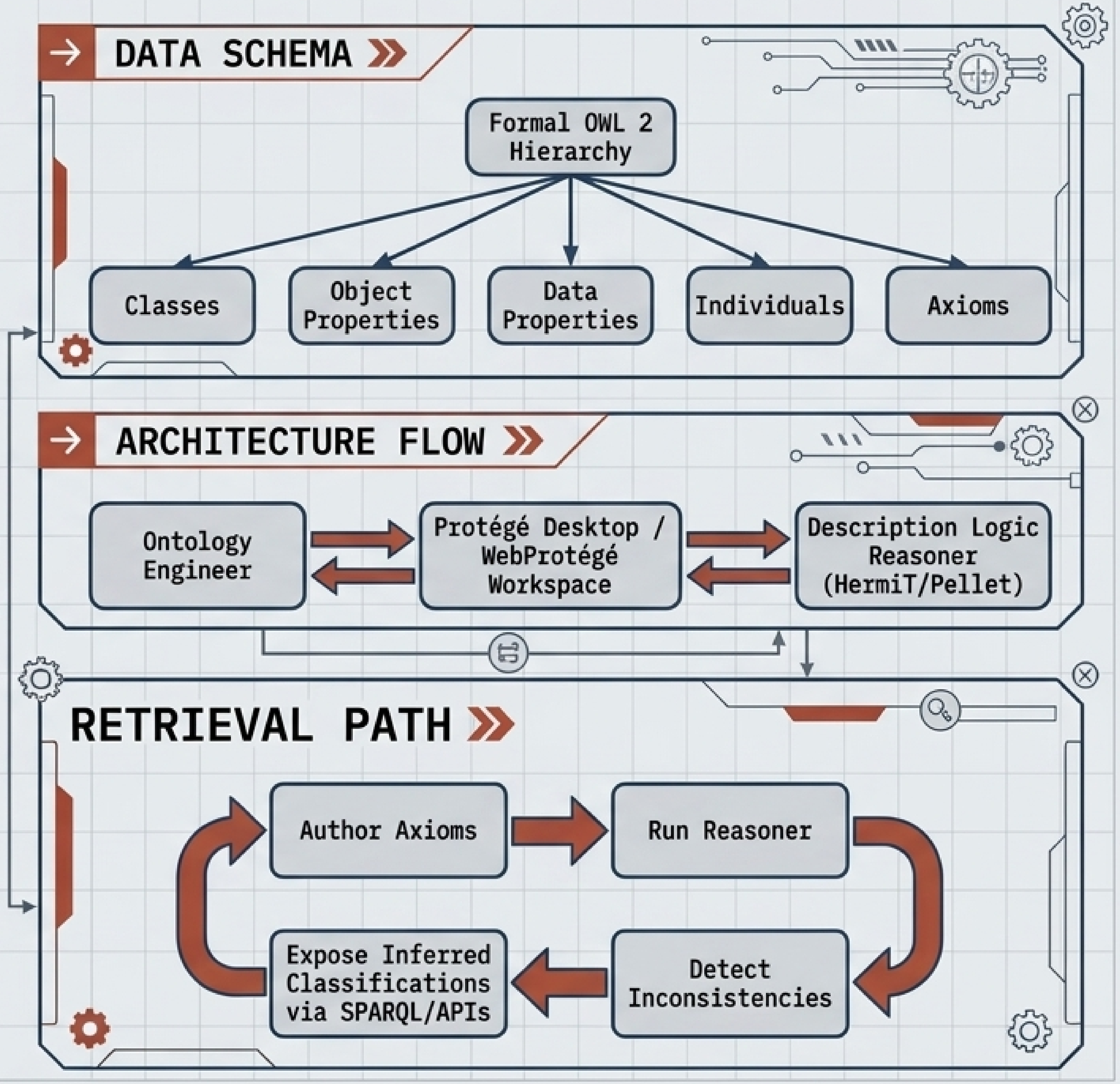
OWL Ontology Engineering Environment

TARGET VECTOR

Complex domains require explicit, machine-readable conceptual models validated for logical consistency before publication.

SIGNATURE MECHANISM

Formal ontology authoring coupled with logic reasoner feedback and collaborative revision tracking.



SUPERMEMORY

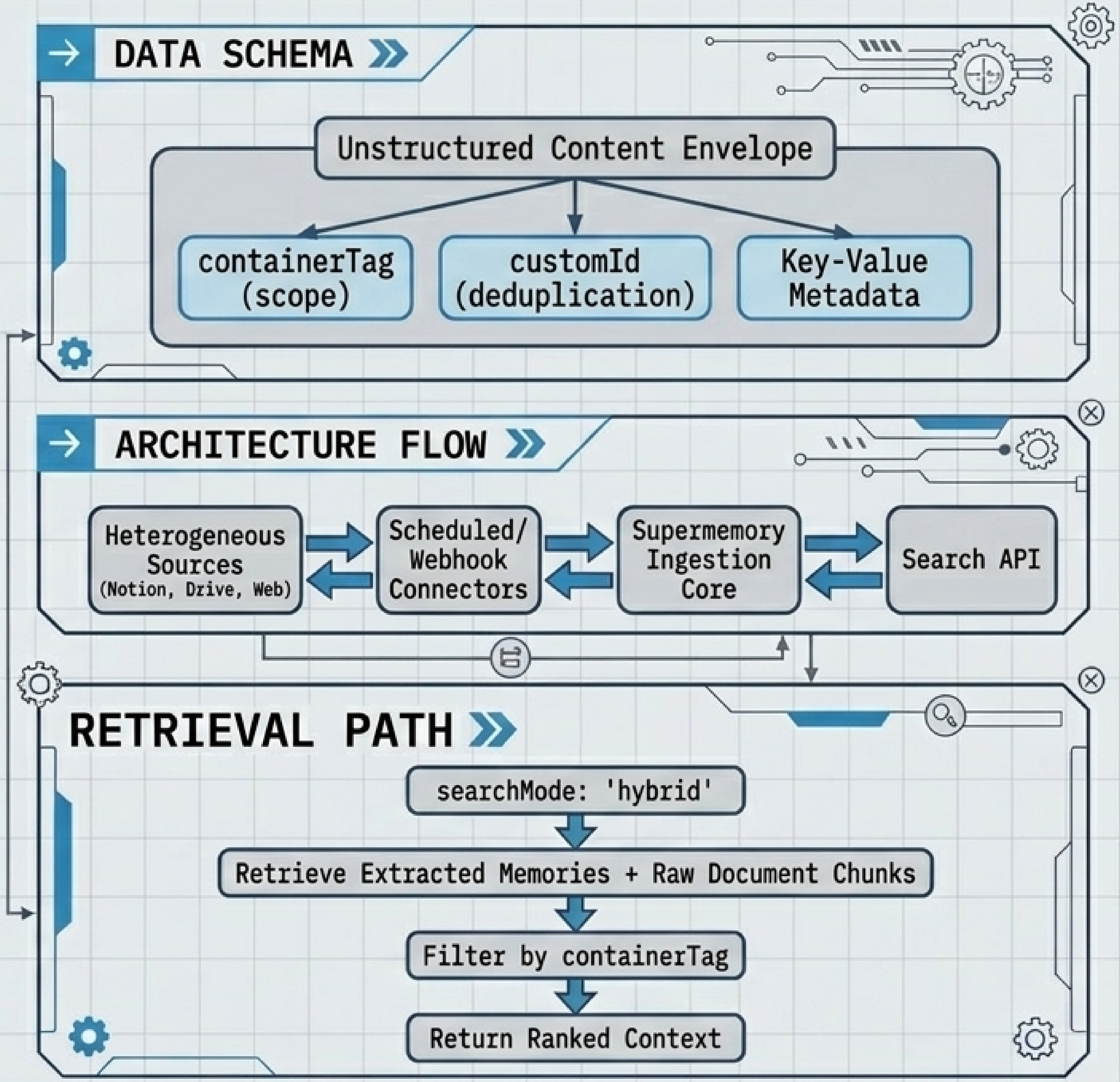
Universal Memory and Context API

TARGET VECTOR

Agent context is scattered across heterogeneous sources; teams need one endpoint to process inputs without custom pipelines.

SIGNATURE MECHANISM

Universal format ingestion feeding a centralized memory extraction engine.



TOPBRAID EDG

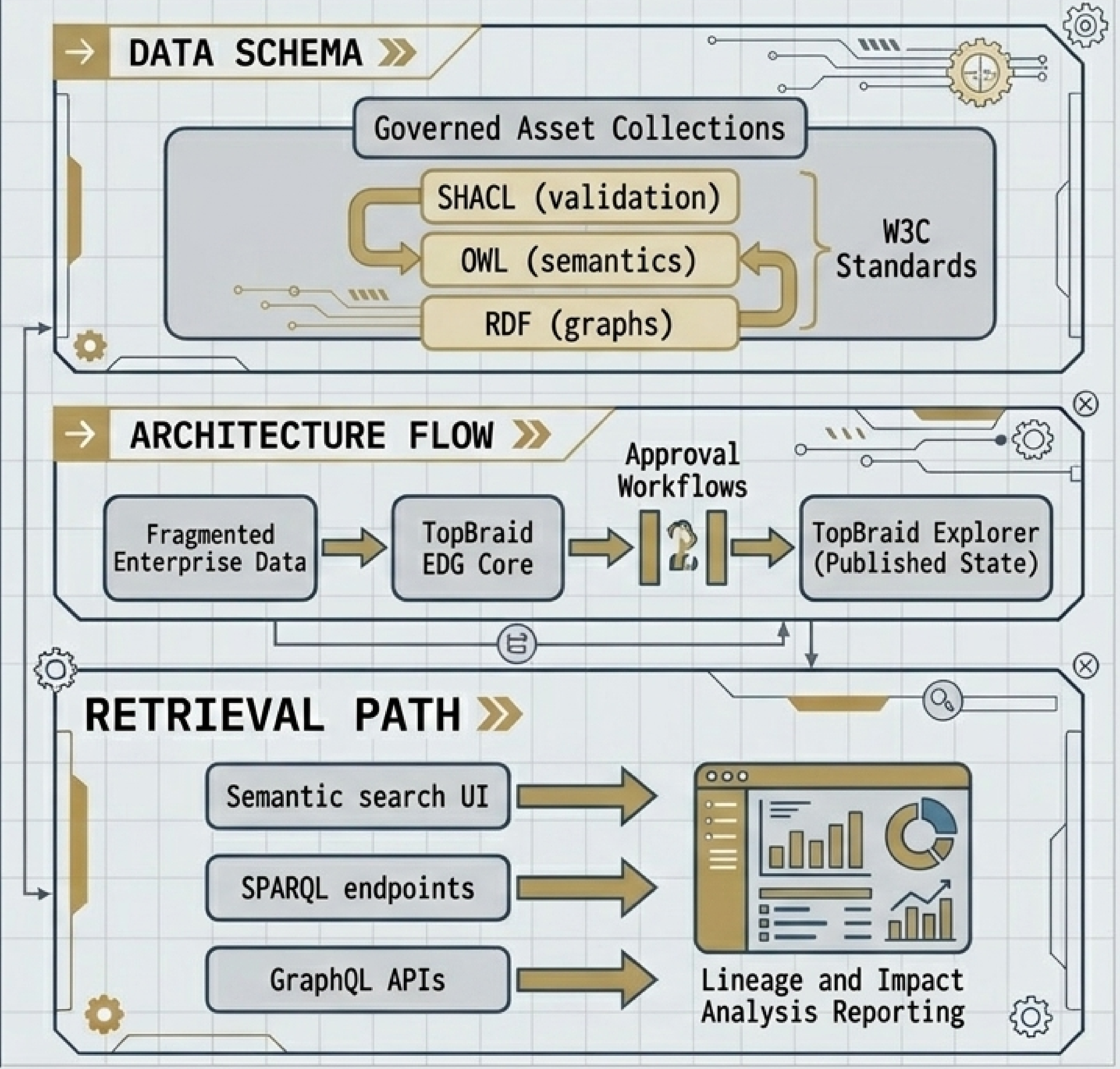
Enterprise Semantic Data Governance Platform

TARGET VECTOR

Large organizations lack an auditable mechanism to define terms, lineage, and ownership across across fragmented systems.

SIGNATURE MECHANISM

Governed semantic asset management through isolated working copies, peer reviews, and strict approval workflows.



ZEP (GRAPHITI)

Temporal Knowledge Graph Memory

TARGET VECTOR

Facts change over time. Flat vector databases retrieve stale information, erasing the chronological evolution of data.

SIGNATURE MECHANISM

Continuous extraction of dynamic facts connected by explicit temporal edges tracking validity intervals.